

PH.D. QUALIFYING EXAM 2021: STATISTICAL INFERENCE

Rules: This is a four hour exam consisting of two problems, both of which should be answered. You can take the exam at any location where you have the facility to scan or take pictures of your answers and convert them into pdf files. You are allowed to use any material of your choice prepared before the time of the exam, but use of online materials or consulting anyone else during the exam is not be permitted. If you have any questions during the exam, please email Professors Harry Crane (hcrane@stat.rutgers.edu) and Tirthankar Dasgupta (td370@rutgers.edu).

Please make sure to **handwrite** your answers clearly and legibly on blank papers, preferably using pens, leaving an impression deep enough for the papers to be scanned. **Typed answers are not permitted.**

Create two pdf files for your answers - one for each problem. Both the pdf files containing your answers should be emailed to td370@stat.rutgers.edu and hcrane@stat.rutgers.edu by 2:15 PM (or at an earlier time based on your location as agreed upon in your communication with the Graduate Directors). **Show your work and justify your answers clearly. If you use known results (e.g., results stated/proven in class), please state them clearly.** Good luck!

Problem 1

Instructions:

- If you cannot solve/prove one claim completely, provide as many partial details as possible.
- Your final score for this problem = $\min\{100, \text{raw score}\}$.

Part I. (75 points)

- (1) Let $\{P_n\}, \{Q_n\}$ be a sequence of probability measures with P_n, Q_n defined on a measurable space $(\Omega_n, \mathcal{A}_n)$.
 - (a) State the definition for $\{P_n\}$ to be contiguous with respect to $\{Q_n\}$, denoted as $P_n \triangleleft Q_n$.
 - (b) Let $\chi^2(P||Q) \equiv \mathbb{E}_Q((dP/dQ - 1)^2)$ be the chi-squared divergence between P and Q . Show that $P_n \triangleleft Q_n$, assuming $\sup_n \chi^2(P_n||Q_n) < \infty$.
- (2) Let $(P_\theta : \theta \in \Theta)$ be a parametric model indexed by some open set $\Theta \subset \mathbb{R}^k$.
 - (a) State the definition for $(P_\theta : \theta \in \Theta)$ to be local asymptotically normal (LAN) (at θ).
 - (b) Verify the LAN condition provided that $(P_\theta : \theta \in \Theta)$ is ‘smooth’ enough (you need to specify the smoothness conditions).
- (3)
 - (a) State the Neyman-Pearson lemma.
 - (b) Given $\theta_0, \theta_1 \in \mathbb{R}^p$, compute the minimum of the sum of type I and type II errors for the testing problem $H_0 : \theta = \theta_0$ vs $H_1 : \theta = \theta_1$, when the observation is from a p -dimensional Gaussian location model $\{\mathcal{N}(\theta, I_p) : \theta \in \mathbb{R}^p\}$.

Part II. (75 points)

Let P, Q be two probability measures defined on a measurable space $(\mathcal{X}, \mathcal{A})$, and $P_n = P^{\otimes n}$ and $Q_n = Q^{\otimes n}$ be the natural n -fold product measures (defined on the product space $(\mathcal{X}^n, \mathcal{A}^n)$). Suppose (for simplicity) $D_{\text{KL}}(P||Q) < \infty$. Here $D_{\text{KL}}(P||Q)$ is the Kullback-Leibler divergence between P and Q . A test Ψ is defined as a measurable map $\Psi : \mathcal{X}^n \rightarrow \{0, 1\}$. Define for any $\alpha \in (0, 1)$

$$R_\alpha \equiv \liminf_{n \rightarrow \infty} \left[-\frac{1}{n} \log \inf_{\Psi : \mathcal{X}^n \rightarrow \{0, 1\}, \mathbb{E}_P \Psi \leq \alpha} \mathbb{E}_Q(1 - \Psi) \right].$$

A standard calculus exercise shows that R_α can be equivalently characterized as the largest number $r > 0$ such that there exists some integer $n_0 \in \mathbb{N}$, such that if $n \geq n_0$, we may find some (test) $\Psi : \mathcal{X}^n \rightarrow \{0, 1\}$ so that type I error $\mathbb{E}_P \Psi \leq \alpha$, and type II error $\mathbb{E}_Q(1 - \Psi) < e^{-nr}$ (You do not need to prove this, but in case you do not understand it immediately, give it a bit more thought).

- (1) Using Neyman-Pearson lemma, show that $R_\alpha \geq D_{\text{KL}}(P||Q)$.
(Hint: Prove that for any fixed $\varepsilon > 0$, $R_\alpha \geq D_{\text{KL}}(P||Q) - \varepsilon$)

- (2) Prove the inequality: for any event A and $x > 0$,

$$P(A) - x \cdot Q(A) \leq P\left(\log \frac{dP}{dQ} > \log x\right).$$

(*Hint:* Write the left hand side in integral form)

- (3) Choose appropriate $x > 0$ in (2) to conclude the reversed inequality $R_\alpha \leq D_{\text{KL}}(P||Q)$.

Problem 2

Consider the following regression problem. Let $y_1, \dots, y_n$ be jointly normal with $E(y_i) = \beta x_i$ and $x_1, \dots, x_n$ being known constants, $\text{var}(y_i) = \sigma^2$ for $i = 1, \dots, n$, and $\text{corr}(y_i, y_j) = \rho$ for $i, j = 1, \dots, n$ with $i \neq j$. The parameters satisfy the conditions $\beta \in \mathbb{R}$, $\sigma^2 > 0$, and $-1/(n-1) < \rho < 1$.

1. Consider the case of known ρ with (β, σ^2) being unknown parameters. If it exists, find the minimum variance unbiased estimator of β among all linear estimators, i.e. among estimators of the form $\hat{\beta} = a_1 y_1 + \dots + a_n y_n$. *If it does not exist, justify why not.*
2. Consider the case of known β with (σ^2, ρ) being unknown parameters. Find the maximum likelihood estimator for (σ^2, ρ) . *Clearly justify why it is the mle.*
3. Consider now the case for which (β, σ^2, ρ) are all unknown parameters.
 - a) Find the minimal sufficient statistic, and find necessary and sufficient conditions on $x_1, \dots, x_n$ for the minimal sufficient statistic to be complete. *Clearly justify your answer.*
 - b) If they exist, find the maximum likelihood estimator for (β, σ^2, ρ) for the following two cases: i) $x_1 = \dots = x_n$, and ii) $x_1 + \dots + x_n = 0$.
If the maximum likelihood estimator does not exist, justify why not.

Note

A useful result from linear algebra: Let $\mathbf{I}_n$ be the identity matrix of order n , $\mathbf{a} \in \mathbb{R}^n$ and $b \neq -1/\mathbf{a}^T \mathbf{a}$, then $(\mathbf{I}_n + b \mathbf{a} \mathbf{a}^T)^{-1} = (\mathbf{I}_n + c \mathbf{a} \mathbf{a}^T)$ for some $c \in \mathbb{R}$.